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Structural rotation and volcanic source implications of magnetic data from Eocene volcanic rocks, SW Idaho

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Abstract

Eocene crystal-rich ash-flow tuffs in the southwest part of the Challis volcanic field in Idaho yield a biotite K–Ar isochron date near 45.38 ± 0.19 Ma. This age determination is in good agreement with a previous result [1] from this locality, near Poison Creek, which corrected for modern decay constants is near 44.7 ± 0.8 Ma. AF and thermal demagnetization experiments on 113 oriented cores from sixteen sites in these Challis ash-flow tuffs provide a tilt-corrected mean paleomagnetic direction with $I = +63.6°$, $D = 019.4°$, $\alpha_{95} = 6.2°$, $N = 15$, $k = 39.4$. The corresponding paleomagnetic pole (74.9°N, 313.3°E, $A_{95} = 8.9°$) lies clockwise 34.0° ± 9.8° relative to a synthetic 40 Ma reference pole for North America [2]. The observed inclination (63.6°) is close to that expected (61.8°) for this locality and age. Therefore, a clockwise tectonic rotation of about 34° is inferred to have affected these rocks in the interval 40 to 15 Ma, approximately, accompanying basin-and-range regional extension. AMS analyses of these ash-flow tuffs yield a predominant E–W trend for the maximum susceptibility axis. Assuming that this axis reflects the primary flow fabric of the rock, and that this direction was also rotated by the amount indicated in the paleomagnetic results, a flow source along a trend N55°E–S55°W is calculated. A source region northeast of the study area, in the main region of development of the Challis volcanic field is consistent with these results. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: rotation; paleomagnetism; Eocene; Idaho; ash-flow tuff; geochronology

1. Introduction

The distribution in space and time of volcanic rocks in the Great Basin has become well known following the early work of Armstrong [3] and of Stewart and Carlson [4]. They documented a southward-migrating 'wave' of Cenozoic volcanism in Nevada, culminating in voluminous mid-Cenozoic caldera eruptions in the southern Great Basin. The space–time patterns of volcanic activity for the whole of the North American Cordillera during the Cenozoic are summarized in [5], and for the widespread ash-flow deposits are summarized in [6].

The focus of the present study is Eocene ash-flow tuffs of SW Idaho. In contrast to the widespread silicic tuffs of Oligo–Miocene age in the central Great Basin, older Cenozoic volcanic rocks have more restricted distributions. Outcrops of Upper Eocene calc-alkalic volcanic rocks of the NE Nevada volcanic field [7] extend into west-central Utah, representing volcanic products for the interval from about

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43 to 39 Ma. Approximately coeval volcanic rocks of central and southern Idaho are generally referred to as the Challis volcanic field, and those in NW Wyoming as the Absaroka volcanic field.

In Idaho, Eocene volcanic rocks are assigned to the Challis Volcanic Group (Fig. 1A; [8,9]), replacing the earlier 'Challis Volcanics' [10]. The Challis Volcanic Group is made up of lava, flow breccia, and pyroclastic rocks, with calc-alkaline, peraluminous affinities [11]. K/Ar age determinations for the group are mainly Mid-Eocene, 45 to 51 Ma [1,11–13], with local activity to 39 Ma. Compositions range widely from Mg-rich basalt to alkali rhyolite, but intermediate compositions (dacite and rhyodacite) predominate. Through central Idaho, numerous mappable subunits of the Challis have been recognized and many caldera complexes identified. The Challis Volcanic Group is widely altered and mineralized [13–15].

Here we report results of paleomagnetic, anisotropy of magnetic susceptibility, structural, petrographic and K–Ar geochronologic studies on a small region of the Challis Volcanic Group 30 km southwest of Grand View, Idaho (Fig. 1B) in the vicinity of the basin of Poison Creek (NB: there are several different Poison Creeks in Idaho). Our new age data support the correlation [16] of these rocks with the Challis Volcanic Group. The paleomagnetic data indicate that the area sampled has undergone a significant clockwise rotation. Correcting for this rotation provides a NE–SW orientation to the axis of flow, consistent with a Challis source to the northeast, as inferred from the trends of axes of maximum anisotropic magnetic susceptibility described below.

2. Geology and field methods

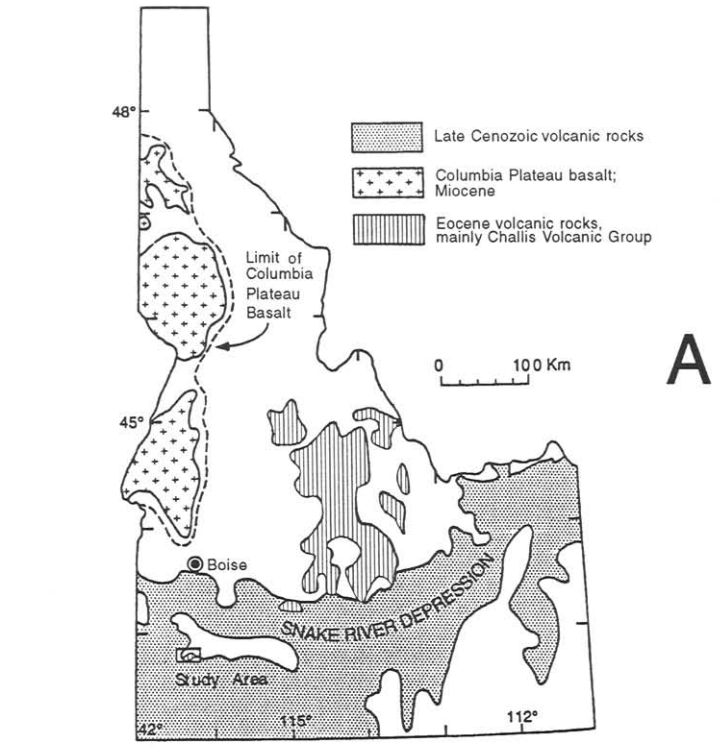
These Challis rocks are crystal-rich, densely welded ash-flow tuffs [16]. Their correlation is based in part on a K–Ar biotite date of 43.6 ± 0.8 [1] from a sample of rhyodacitic welded tuff north of our sampling sites 9 and 15 (Fig. 1B).

In these rocks crystals and crystal fragments are more abundant than lithic or collapsed pumice fragments. The crystals have the following modal ranges [16]: quartz (10–15%), alkali feldspar (7–12%), plagioclase (45–55%), biotite (10–15%), hornblende (5–10%), and other minerals (0–2%). Electron microprobe analyses of thin sections from sites 3 and 6 indicate that the plagioclase is in the compositional range An_{42}–An_{32}. Sanidine is in a more restricted range of Or_{71}–Or_{73} but with up to 4% of the celsian end-member. Whole rock chemical analyses of samples from sites 3, 9, and 12, selected to be free of obvious lithic fragments, indicate a restricted range of compositions spanning the andesite–dacite boundary [17]. Inasmuch as K_2O exceeds Na_2O in all three analyses, these rocks can be further classified as high-K andesites and high-K dacites [18].

From six to nine independently oriented core samples (median of 7) were collected at each of sixteen sites (Fig. 1B). Near the northern contact with older granitic rocks, the rocks are hydrothermally altered; our sampling avoided hydrothermally altered regions to the extent that we could recognize them. Azimuthal components of orientation were made with a sun compass, at almost all sites. For each site, typically five or six attitudes of eutaxitic structures (mainly variably compacted flow foliation) were recorded and averaged for the tilt correction of the paleomagnetic vectorial and AMS axial data. Foliation measurements were also taken at widely scattered points over the sampling region to determine regional structural trends. Gentle to moderate SSE dips prevail.

3. Geochronology

The K–Ar biotite date of Neill [1], corrected for the constants adopted in 1977 [19] corresponds to 44.7 ± 0.8 Ma. To augment this earlier age estimate, we determined a K–Ar isochron date based on analyses of samples from sites 3, 9 and 12. Crushed samples between 20 and 100 mesh were split into two portions, one for argon analysis and the other for potassium analysis. An initial biotite concentrate from the Frantz isodynamic magnetic separator was further improved by heavy liquid separation. Potassium contents were determined by flame photometry with a lithium internal standard on biotite aliquots. Argon was measured by the isotope dilution technique. A second set of analyses was obtained from biotite concentrates further purified by electrostatic techniques. The analytical data and conventionally calculated ages are given in Table 1.



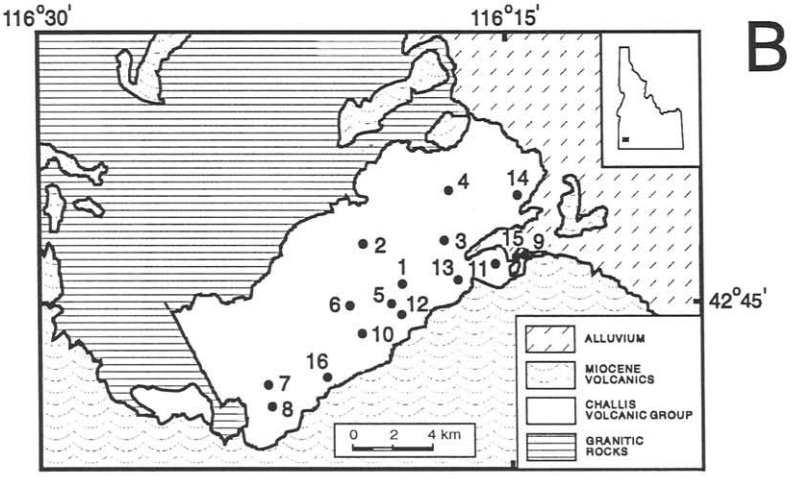


Fig. 1. (A) Map of Idaho showing the study area and the distribution of the Challis Volcanic Group and equivalents in relation to younger Cenozoic rocks of the Snake River depression and of the Columbia plateau-basalt field. (B) Geological map showing sample sites in the drainage basin of Poison Creek in the Challis Volcanic Group of southwest Idaho. Map simplified from Ref. [16].

Table 1
Isotopic age determinations and K–Ar analytical data

Sample No.	Concentration	K (%)	^{40}Ar rad (10^{-6} cm^3/g)	^{40}Ar/^{36}Ar	^{40}K/^{36}Ar ($\times 10^5$)	Age (Ma)
ID-3	Initial	6.165	10.734	2965.6	10.257	44.3
ID-3	Second	6.736	11.739	3427.8	12.026	44.3
ID-9	Initial	5.285	9.029	2087.1	7.018	43.4
ID-9	Second	6.084	10.469	2229.8	7.521	43.7
ID-12	Initial	4.899	8.303	1550.9	4.957	43.1
ID-12	Second	5.307	8.933	1609.4	5.223	42.8

A six-point isochron was calculated from the above data: ^{40}Ar/^{36}Ar intercept = 220; slope = 0.002670 ± 0.0000109; age = 45.38 ± 0.19 Ma (1σ).

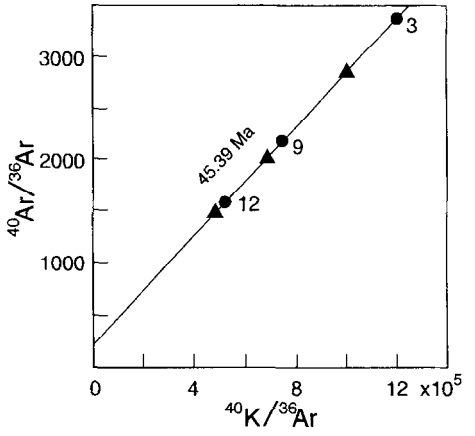


Fig. 2. ^{40}Ar/^{36}Ar vs. ^{40}K/^{36}Ar for samples 3, 9, and 12 from the Challis Volcanic Group. Dots (triangles) represent the second (first) biotite concentrate from each of the samples. The slope of the isochron corresponds to an age of 45.38 ± 0.19 Ma.

These data confirm that the age reported earlier [1], based on a single sample, is representative of that for the Challis Volcanic Group in this region. A more precise and we believe a more accurate age estimate is obtained when the data are analyzed in the form of an isochron (Fig. 2; [20]). The isochron age estimate is 45.38 ± 0.19 Ma (1σ). This date and the conventionally calculated dates (Table 1) are consistent with those from the main outcrop area of the Challis Volcanic Group. Rocks assigned to this group span the interval 51 to 45 Ma in the Challis Quadrangle and in the Lost River and Lemhi Ranges of central Idaho [21,22].

4. Magnetic studies

From each oriented sample, one or more standard paleomagnetic specimens was prepared. Subsequent to measurement of natural remanent magnetization (NRM) but before demagnetization (Table 2), the anisotropy of magnetic susceptibility (AMS) was measured employing a Sapphire Instruments SI-2 low-field instrument. A four repeat measurement of each of six orientations was used on each of 198 specimens. Measured volumes and masses for ideally cylindrical specimens were used to calculate dry weight densities, the means of which are recorded, together with the AMS results, in Table 3. Site-mean AMS axes and their error ellipses were calculated by the Hext-Jelinek method as implemented by Lienert [23] (and Lienert, pers. commun., 1992).

Progressive demagnetization was performed on 113 specimens from the 16 sites, 50 by thermal demagnetization and the remainder by alternating field (AF) demagnetization. AF demagnetization in 8–12 steps was done up to a maximum field of 100 mT using a Schonstedt GSD-5 demagnetizer and a Schonstedt DSM-1 magnetometer. The thermal demagnetization was done using a large volume furnace (1400 cm^3 sample space) enclosed in a 1.5 m square three-axis field-cancelling array and associated magnetizations were measured using a Molspin magnetometer. Eight heating–cooling cycles followed by remanence measurement were performed up to a maximum temperature of 600°C. The results from each specimen were analyzed by orthogonal vector plots [24] and least-squares line regression [25]. Stable endpoints defined by this analysis were combined to obtain site averages with Fisher statistics.

Table 2
Paleomagnetic site-mean directions for the Challis Volcanic Group, Poison Creek area, SW Idaho, tilt-corrected

Site No.	In situ								Foliation		One stage		Two stage	
	N_A	N_T	N_S	D_1 (°)	I_1 (°)	R	k	α_{95} (°)	Φ (°)	θ (°)	D_2 (°)	I_2 (°)	D_3 (°)	I_3 (°)
ID1A	2	2	4	333.2	54.6	3.989	268	5.6	128	33	039.0	75.7	037.2	75.7
ID1B	1	1	2	025.7	54.3	1.995	196	17.9	155	20	056.2	62.8	054.7	62.8
ID2	4	2	0											
ID3	5	3	8	006.6	51.2	7.977	308	3.2	127	14	024.6	56.4	023.8	56.4
ID4	5	3	8	035.0	57.4	7.978	317	3.1	186	11	046.9	66.5	046.0	66.5
ID5	3	3	6	341.1	47.0	5.994	777	2.4	142	32	016.5	74.1	014.3	74.1
ID6	3	3	6	020.5	68.8	5.973	185	4.9	142	14	059.8	71.9	058.9	71.9
ID7	4	3	5, 2	314.1	67.7		130	5.9[a]	020	30	349.9	46.4	352.0	46.4
ID8	4	2	5	327.9	68.1	4.983	235	5.0	043	25	006.0	53.2	007.3	53.2
ID9	4	4	8	007.2	67.9	7.950	140	4.7	008	07	007.4	60.9	007.9	60.9
ID10	3	3	6	009.4	67.5	5.991	587	2.8	015	13	011.3	54.5	012.3	54.5
ID11	5	3	5, 1	007.8	55.0		110	7.0[a]	156	11	019.2	63.8	018.3	63.8
ID12	3	4	7	352.4	42.7	6.963	161	4.8	139	13	001.3	53.0	000.4	53.0
ID13	3	4	7	026.7	46.3	6.986	445	2.9	173	23	050.2	62.9	048.4	62.9
ID15	6	3	9	347.7	69.0	8.973	300	3.0	054	11	008.2	62.7	008.7	62.7
ID16	4	3	3, 2	343.7	70.1		27	17.9[a]	062	11	008.6	65.5	008.9	65.5
Mean (in situ)			15	000.3	61.1	14.516	29	7.2						
Mean (one-stage tilt)			15			14.644	39	6.2			019.4			
Mean (two- stage tilt)			15			14.661	41	6.0					019.2	63.5

N_A is No. of specimens treated by A.F. demagnetization; N_T is No. of specimens thermally demagnetized; N_S is No. of specimens yielding stable endpoints (SEPs). A second entry (sites 7, 11 and 16) is the No. of specimens defining demagnetization great circles which were combined with the SEPs to yield the site-mean direction. D_1 and I_1 are in situ declination and inclination, respectively; D_2 and I_2 are declination and inclination after simple tilt correction; D_3 and I_3, corrected for fold-axis plunge and bedding tilt. R, k and α_{95} are Fisher [27] statistical parameters; superscript 'a' is the square root of the product of the minimum and maximum radii of the 95% confidence ellipse [26]. Φ and θ are azimuth of dip and dip angle, respectively, for tilt correction.

Table 3
Site-mean AMS parameters and densities, Challis Volcanic Group, SW Idaho

Site	ρ	S.d.	K	K_1/K_2	K_2/K_3	K_1			K_2			K_3		
						D (°)	I (°)		D (°)	I (°)		D (°)	I (°)	
ID1	2.46	0.09	18.2	1.003	1.020	275	10	(64,22)	008	16	(64,24)	154	71	(31,17)
ID2	2.30	0.10	2.51	1.003	1.010	300	23	(42,11)	209	03	(41,17)	111	66	(28,12)
ID3	2.67	0.10	18.0	1.008	1.020	247	12	(12,03)	155	08	(12,06)	032	75	(07,03)
ID4	2.59	0.03	14.4	1.009	1.004	252	15	(15,05)	349	23	(39,14)	132	62	(39,07)
ID5	2.55	0.05	17.7	1.006	1.014	284	02	(34,07)	014	12	(35,11)	183	78	(13,06)
ID6	2.53	0.02	16.8	1.003	1.009	264	42	(33,20)	052	44	(33,14)	158	16	(21,11)
ID7	2.57	0.04	14.2	1.017	1.011	077	04	(10,05)	167	03	(25,09)	300	85	(25,06)
ID8	2.55	0.03	16.2	1.007	1.007	085	06	(16,13)	177	18	(24,13)	339	71	(23,14)
ID9	2.52	0.05	18.0	1.004	1.012	265	08	(42,14)	173	14	(41,14)	024	74	(18,13)
ID10	2.53	0.03	18.7	1.010	1.007	268	01	(18,13)	177	23	(30,16)	000	67	(29,13)
ID11	2.55	0.04	19.0	1.000	1.026	105	03	(87,05)	196	16	(87,06)	006	73	(09,05)
ID12	2.53	0.03	11.0	1.005	1.018	344	06	(33,08)	074	01	(33,09)	172	84	(14,05)
ID13	2.54	0.03	10.8	1.003	1.021	272	12	(50,06)	004	08	(50,09)	126	76	(10,07)
ID14	2.32	0.08	2.67	1.005	1.013	331	10	(19,12)	241	03	(20,18)	134	80	(20,11)
ID15	2.45	0.04	16.0	1.005	1.026	204	09	(27,06)	113	05	(27,05)	356	80	(06,06)
ID16	2.59	0.03	16.0	1.008	1.005	266	31	(22,15)	013	27	(33,22)	136	47	(33,15)

5. Paleomagnetic results

Initial NRM intensities of most samples from the collection are in the 1–10 A m^{-1} range; samples from sites 2 and 14 have intensities which are about a decade lower. The majority of samples from the collection exhibit simple demagnetization patterns (Fig. 3A,B) with characteristic magnetizations directed north and down. A few samples exhibited low coercivity components of diverse orientations that were removed by 40 mT in AF treatment or at 400–500° of thermal demagnetization after which the characteristic direction is isolated (Fig. 3C,D). Site 2, in scarce sedimentary interbeds on the edge of the hydrothermally altered zone, yielded weak and unstable remanent magnetizations. Results from site 14 are anomalous from many points of view: structural orientation, remanence direction, and AMS pattern. Both sites 2 and 14 appear to be hydrothermally altered judging from the bleached color of the samples

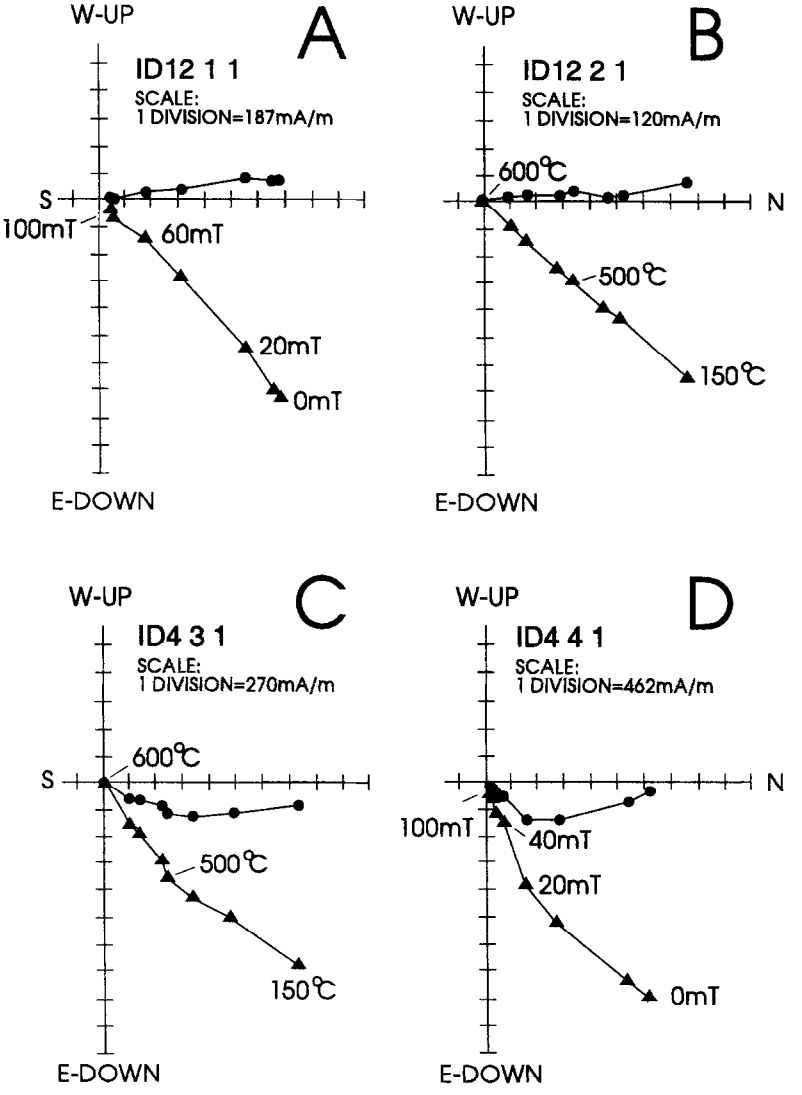


Fig. 3. Orthogonal vector diagrams for pairs of specimens from sites 4 and 12 examined under AF (A, D) and thermal (B, C) demagnetization. Both demagnetization techniques reveal a magnetization down to the north or northeast as the characteristic remanence.

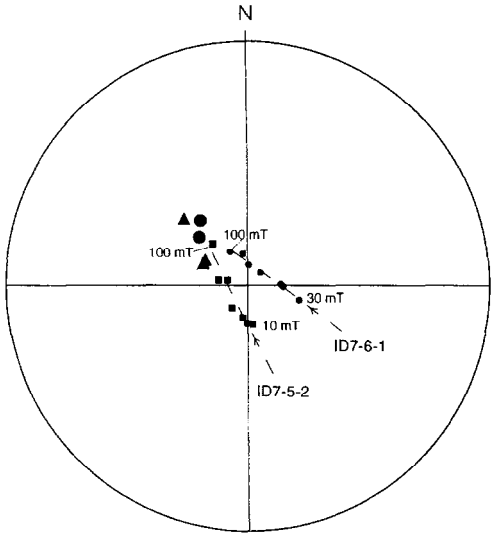


Fig. 4. Equal-area projection for site ID-7 showing five stable endpoint directions obtained by AF (large circles) and thermal (large triangles) demagnetization. Two specimens fail to yield characteristic directions under AF demagnetization, but converge along great circles, marked by small circles and small squares, towards the characteristic direction for the site.

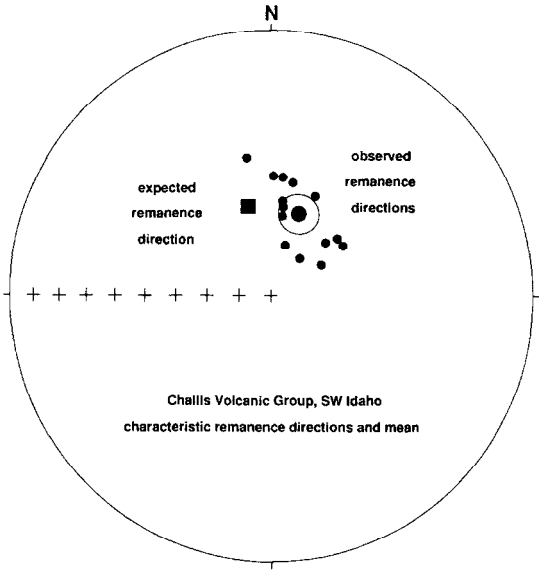


Fig. 5. Paleomagnetic site-mean and formation-mean directions, corrected for simple tilt, with α_{95} confidence circle, for Challis Volcanic Group of the Poison Creek region of SW Idaho. Also shown is the expected reference direction computed from a synthetic 40 Ma reference paleomagnetic pole [2].

and from the low bulk susceptibilities (Table 3). Results from these two sites are excluded from further paleomagnetic analysis.

From sites 7, 11 and 16, most samples yielded stable endpoints but the remanence directions for a few migrated toward the stable endpoint without reaching it at the highest AF fields (Fig. 4). At these three sites coherent great-circle demagnetization trajectories were combined with the stable endpoint data [26].

The site-mean characteristic paleomagnetic directions were subjected to Fisher statistical analysis [27]. Formational means were computed in three ways: (1) in situ results, (2) simple tilt-correction and, (3) two-stage: fold-plunge correction followed by tilt-correction. With simple tilt-correction the formational mean direction is defined with higher precision relative to the in situ mean (Table 2). The improvement in clustering is not significant at the 95% confidence level [28,29]. However, the tilts are mainly gentle. By limiting 'unfolding' to 70%, the precision of the formational mean reaches statistical significance under McFadden's test [29]. The site-mean paleomagnetic directions and the formation mean, corrected for simple tilt, are shown in Fig. 5.

When in situ site-mean directions, connected to their fully untilted directions by small circles, are examined in stereographic projections, a maximum precision of directions at partial untilting may arise at the 'crossover' condition (for example, see fig. 6 in [30]). These 'crossover' patterns have been interpreted as evidence for synfolding magnetizations but their geometry can have other explanations. Those include combinations of pre- and post-folding magnetizations unsuccessfully separated by demagnetization techniques, or internal tectonic strain invalidating the simple rigid-body-rotation assumption in the tilt correction. We obtained crossover patterns only for sites 7 and 8. We suggest that some unknown amount of initial dip at those sites accounts for the crossover pattern. Otherwise the generally small amount of scatter in the primary magnetizations can be attributed to paleosecular variation and imperfect tectonic corrections.

To further evaluate the tectonic correction we employed a two-stage tilt correction noting that the poles of normals to foliation form a partial girdle (Fig. 6A). The symmetry or 'fold' axis of this girdle, as determined by eigenvector methods for the mean

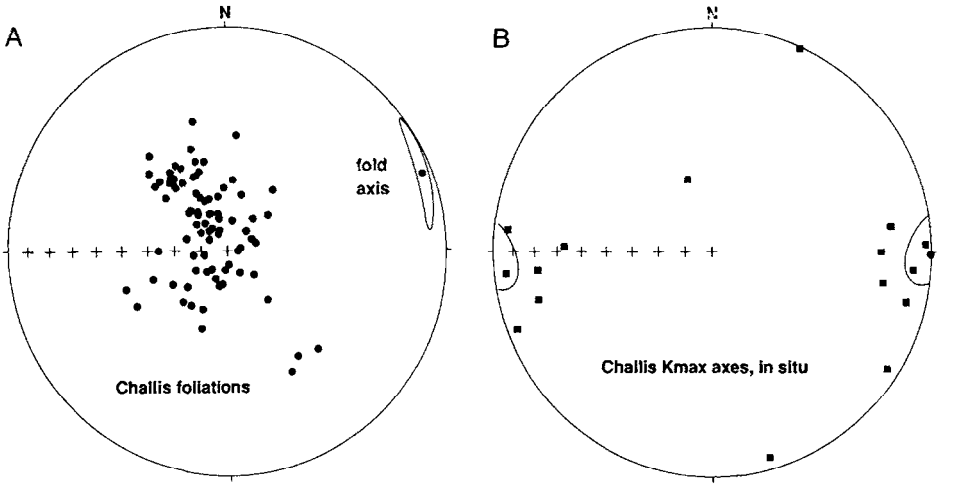


Fig. 6. (A) Equal-area projection of regional foliation trends (84 poles) and the corresponding symmetry ('fold') axis and 95% confidence ellipse of their partial girdle distribution. (B) K_{max} axes in situ; the Bingham mean and its corresponding 95% confidence ellipse are shown for thirteen selected sites of the main cluster.

foliation normals at fifteen sites, plunges 09° towards 083°. This direction was rotated to horizontal and the foliations and characteristic remanent directions similarly rotated, after which the characteristic directions were further corrected for the residual foliation tilt. Mean directions from this procedure have a better clustering ($k = 41$, Table 2) than the in situ mean ($k = 29$). However, the two-step tilt correction improves clustering only marginally relative to a simple one-step tilt correction ($k = 41$ vs. $k = 39$). The two-step correction has the effect of offsetting declinations slightly towards north, by up to a degree or so, while not significantly changing the paleomagnetic inclinations. We conclude that the simple tilt correction is adequate for evaluating directional geometry in these rocks.

The mean inclination (63.6°) differs little from the expected inclination (61.8°) calculated from the synthetic 40 Ma reference pole for North America [2]. However, the mean declination (019.4°) is 34.0±9.8° clockwise from the expected direction (345.4°) implying clockwise tectonic rotation (Fig. 5) as discussed later.

6. AMS results

As for other ash-flow tuffs, the degree of anisotropy [31] is low, 1–3% (Table 3), and fabrics are mainly oblate (Fig. 7). However, two sites (7 and 16) have slightly prolate fabrics. As expected from geometric considerations, the prolate sites have slightly larger scatter of K_{min} axes than most other sites (Fig. 8). In situ site-mean K_{max} axes trend nearly horizontally with an E–W trend (Fig. 6B).

In general, foliation poles and K_{min} axes agree well. Both define a partial girdle distribution reflecting a regional tilt about a symmetry or 'fold' axis, as discussed later. Although rock strain can alter the orientation and axial ratios of AMS ellipsoids [30], the close correspondence here between field measured foliation and magnetic 'foliation' (i.e. the K_{max}–K_f plane) suggests that the magnetic fabrics of these volcanic rocks have not been substantially modified by tectonic strain. Early compaction is evident in many ash-flow tuffs. However, a variable compaction strain, as in the Bishop tuff [32], does not necessarily produce a corresponding effect in the AMS signature [33], possibly due to devitrification magnetic mineralogy which mainly post-dates compaction.

We believe that the mean K_{max} axes for our Challis sites represent primary flow axes associated with deposition of these ash-flow tuffs. As is the case for many massive ash-flow tuffs, no visible field expression of flow lineations (e.g. aligned lithic clasts, crystals, shards, etc.) could be found to compare with the K_{max} orientations.

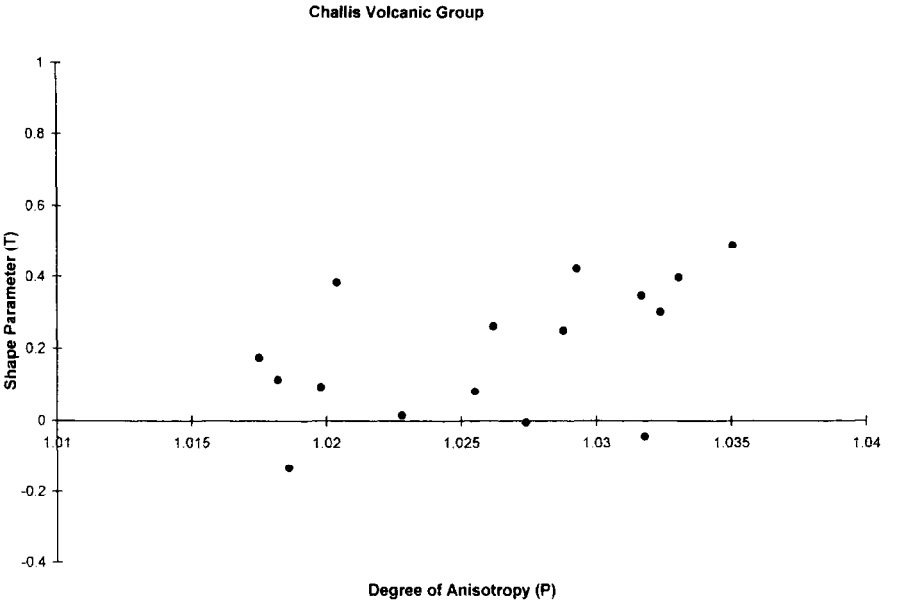


Fig. 7. Shape parameter (T) vs. degree of anisotropy (P) [31] shows dominance of oblate AMS ellipsoids (positive T values) and low overall anisotropy, to about 3.5% (P values).

7. Discussion

The total number of independently erupted flows represented in the 16 sites and the time-span over which they erupted are not known. These are important because they bear on the question of whether or not the observed remanence represents a geomagnetic field direction which has adequately averaged secular variation. The process of devitrification is time-dependent and produces new magnetic oxides, but likewise has unknown temporal parameters. At several sites, two flow units were apparent, in one case separated by a sandstone interval, and three flow units were evident at one site. However, as is the case for many ash-flow deposits, the number of cooling units is difficult to ascertain. Thus although we believe we have sampled many flow units, the number is unknown, as is the lateral extent of units sampled and precise correlation between them. With these uncertainties in mind, the remanence results are interpreted with the assumption that they represent a geomagnetic field direction adequately averaged in time both by spacing of eruptions in time and by devitrification processes through time.

The paleomagnetic results of this study imply a tectonic rotation of the sampling area by 34.0 ± 9.8° clockwise (Fig. 9) relative to a 'synthetic' 40 Ma reference pole for North America [2]. Assuming that associated deformation occurred prior to the deposition of unconformably overlying Miocene volcanic rocks, the rotation occurred approximately between 45 and 15 Ma.

The implied rotation affects other features in the 'direction space' of the rocks sampled, such as petrofabric directions and AMS axes. Because the anisotropy of magnetic susceptibility can reflect secondary aspects of rock fabric such as strain effects, as well as primary aspects, and because the K_{max} axes lie near the symmetry or 'fold' axis of the foliation girdle, the K_{max} distributions were explored in more detail. Bingham statistics on 84 foliations measured over the entire sampling area indicate a regional symmetry axis plunging 05° towards 069° (Fig. 6A). The mean of fifteen in situ K_{max} site-mean axes plunges 0° towards 091° (Fig. 6B). For selected K_{max} axes most clustered near this direction (i.e. for thirteen sites, excluding sites 12, 14, 15), the Bingham mean plunges 01° towards 091°; this direction is insignificantly different from that for all fifteen sites. Thus the K_{max} axes lie about 20° from the symmetry ('fold') axis of the girdle defined by the foliation poles, and probably represent a primary flow

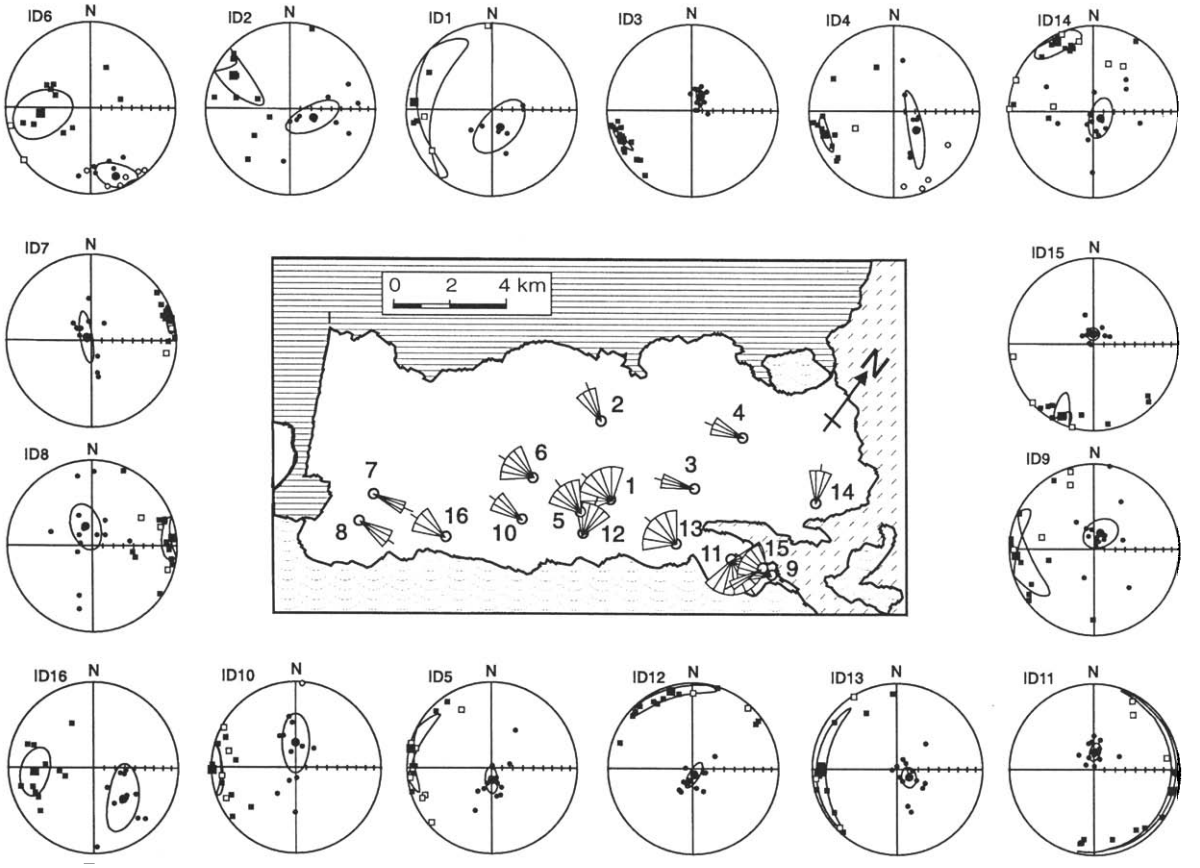


Fig. 8. Equal-area stereograms for each site in which the tilt-corrected axial directions K_{max} and K_{min} are plotted as squares and circles, respectively, together with their associated error ellipses. For each site the horizontal projection of the K_{max} error ellipse has been transferred to the map and represented by a fan the apex of which is at the collecting site.

fabric effect rather than a secondary strain effect.

The relationship between the foliation symmetry 'fold' axis and the K_{max} mean is seen more clearly in a projection which centralizes these features (Fig. 10). This view is effectively an 'upper hemisphere' projection with the unit reference sphere rotated. In this figure, the eastern hemisphere projections of the K_{max} axes are represented. The 95% confidence regions around the Bingham means are elliptical rather than circular (Fig. 10A). The angle between the means is $\alpha \pm \varepsilon$, where ε, the error in the angle between the means, is taken to be

$$\varepsilon = \frac{1}{\sqrt{2}} \left(r_1^2 + r_2^2\right)^{1/2} \quad (1)$$

in which r_1 and r_2 are the error angles of the respective elliptical cones of 95% confidence along the plane joining the Bingham mean directions. For the foliation fold axis and the mean of thirteen selected in situ K_{max} directions, the angle between these directions is $22.0 \pm 12.9°$; relative to the mean of all fifteen in situ K_{max} directions, the angle is $21.9 \pm 15.3°$. In both cases the fold axis and the mean K_{max} direction are significantly different.

Upon correction for simple tilt, the clustering about the K_{max} mean improves, and for the thirteen selected K_{max} directions becomes almost Gaussian, with a conical error near 9° (Fig. 10B). In this tilt-corrected pattern, the K_{max} axes define a plane dipping 11°W, or equivalently, inclined upwards the same amount towards the east as can be inferred from the figure.

The interpretation of these tilt-corrected K_{max} directions is an interesting challenge. If they represent

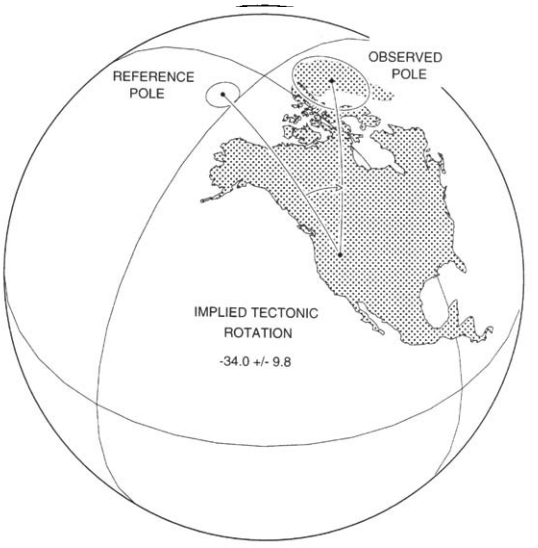


Fig. 9. The paleomagnetic pole for the Challis Volcanic Group of the Poison Creek vicinity indicates a 34° clockwise tectonic rotation relative to a 'synthetic' 40 Ma [2]. Associated directions in these rocks, such as petrofabrics and AMS axes, have been rotated by a similar amount.

primary flow directions as we believe they do, the thirteen tilt-corrected K_{max} axes indicate a mean plunging 11° towards 269°. Correcting this direction for the paleomagnetically indicated rotation of 34° clockwise yields a rotation-corrected primary flow axis along 55°/235°. This is consistent with a source for these Poison Creek ash-flow tuffs to the northeast in the main region of Challis eruptions (see Fig. 1A), or to the southwest. Although a NE source seems reasonable from a regional volcanic perspective, we note that the 11°W plunge of the K_{max} axes is steeper than typical, but favors a source to the southwest. In many ash-flow tuffs, the K_{max} axes plunge 0° to 7° in the source direction [34]. The implication here is that if the source is in the SW direction, these flows are not part of the main Challis volcanic field.

For a NE source, there arises a 'plunge anomaly' here of 11° to 18°W in the tilt-corrected flow-axis. Perhaps this anomaly is partly attributable to deposition of tuffs onto a surface of significant relief. Locally, ash-flow tuffs can flow uphill or across slopes because of velocity-momentum considerations; however the 'flow length' sampled here is at least 10 km long, a seemingly long stretch for such a mechanism. Alternative explanations could involve residual strain effects, measurement errors, or other unknown influences. Two observations suggest that strain effects here are small: (1) the arguments presented above indicate that the foliation symmetry ('fold') axis is significantly different from the mean K_{max} direction; and (2) the mean K_{max} direction (approximately E–W) is clearly oblique to the trend of the principal faults and contacts which trend N70E (see Fig. 1B).

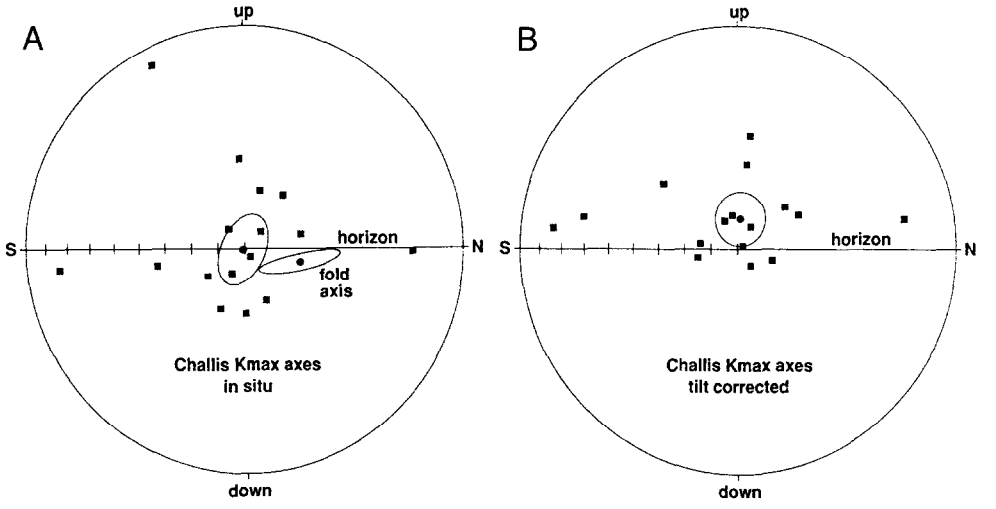


Fig. 10. (A) Comparison of Bingham means and 95% confidence ellipses for the foliation symmetry ('fold') axis and mean K_{max} axis for selected sites, in situ (the mean excludes sites 12, 14, 15). The projection is an external view of the unit projection sphere, with all data projected into the eastern hemisphere; view is towards the west. (B) A similar projection showing the K_{max} axes, after correction for simple tilt, lying in a plane dipping 11°W (depicted here in the eastern hemisphere as dipping upwards to the east).

Finally we observe that, upon making the tilt-corrections for the Challis ash-flow tuffs, the K_{min} directions display a residual small N–S girdle. This pattern appears to be unrelated to the symmetry of the foliation poles. Possibly this K_{min} distribution is influenced by small components of initial dip inward towards the E–W 'channels' of flow of the tuffs.

8. Conclusions

The principal results of this study are that:

(1) K–Ar isotopic data support an age near 45 Ma for the ash-flow tuffs of the Challis Volcanic Group in the vicinity of Poison Creek in SW Idaho;

(2) paleomagnetic evidence suggests that these rocks were rotated tectonically about 34° clockwise in the interval between 45 and 15 Ma, approximately;

(3) AMS analyses are consistent with an axis of flow presently oriented along an E–W axis;

(4) the AMS flow axis, corrected for tectonic rotation of 34° clockwise, trends N55°E–S55°W from the Poison Creek region, possibly indicating a source towards the main area of eruption of the Challis Volcanic Group.

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